

AI Invoice Processing

An n8n pipeline that turns invoices and receipts from Gmail or Drive into validated, categorised, approved ledger entries — with Slack handling both error alerts and interactive approvals.



1 Executive Summary

Manually keying in invoices is slow and error-prone, and a purely automated pipeline risks posting bad data straight to the books. This workflow strikes a middle ground: every document is read by AI, checked against basic arithmetic and required fields, deduplicated against the ledger, auto-categorised, and then either logged straight through or routed to a human via interactive Slack buttons — so the business gets speed on the easy cases and a real approval trail on the ones that matter.

2 Key Metrics

Intake Channels	Gmail (unread invoice emails with attachments) and a watched Google Drive folder
Extraction Engine	Google Gemini, reading PDFs and images directly into structured invoice JSON
Validation	Required fields present, and subtotal + tax must reconcile with the stated total
Duplicate Protection	The ledger is searched by invoice number before anything new is logged
Categorisation	Gemini assigns one of six expense categories from vendor and line items
Approval Threshold	Invoices up to \$500 auto-approve; anything higher needs a Slack decision
Approval Timeout	An unanswered Slack approval resolves to “No Response” after 3 days

\$500

AUTO-APPROVE THRESHOLD

3 days

APPROVAL TIMEOUT

2

INTAKE CHANNELS

3 Business Value

- ✓ **No manual data entry** — Gemini reads the PDF or photo directly and returns structured, ledger-ready fields.
- ✓ **Bad extractions caught before they're trusted** — arithmetic and required fields are validated before anything is logged.
- ✓ **No duplicate invoices** — every document is checked against the ledger by invoice number before it's processed.
- ✓ **Spend control without bottlenecking** — routine invoices post automatically, larger ones get a real Slack decision.

4 Architecture Overview

Two independent triggers feed one shared processing pipeline, plus a dedicated error-handling branch.

FLOW	TRIGGER	PURPOSE
Email Intake	Gmail trigger, polling every minute	Picks up unread emails matching invoice/receipt/bill with an attachment
Drive Intake	Google Drive trigger, on new file	Picks up any file dropped into a designated invoices folder
Shared Pipeline	Fed by either intake branch	Extraction, validation, dedup, categorisation, approval routing and ledger logging

5 How It Works

Intake & Extraction	Categorise, Approve & Log
1 A Gmail trigger polls for unread invoice emails with an attachment; a Drive trigger watches a folder. 2 Each file's MIME type is checked — non-PDF/image files are flagged to Slack for manual handling. 3 Gemini reads the document directly and returns structured JSON: vendor, invoice number, dates, total. 4 Required fields are checked and the subtotal + tax is verified to reconcile with the stated total. 5 The ledger is searched by invoice number; a match skips the invoice with a Slack notification.	1 A second Gemini call assigns one of six expense categories from the vendor and line items. 2 Invoices at or under \$500 are marked Approved and logged straight to the ledger. 3 Anything above the threshold is marked Pending Review and logged before any approval request. 4 An interactive Slack Approve/Reject message is posted; an unanswered request times out after 3 days. 5 The clicked outcome (or timeout) becomes Approved, Rejected or No Response, and the ledger is updated.

6 Reliability & Security Design

- ◆ **Extraction is never trusted blindly** — required fields and basic arithmetic are checked before an invoice is valid.
- ◆ **Duplicate-safe by design** — every invoice is looked up by invoice number before it's processed further.
- ◆ **Spend-gated approvals** — the \$500 threshold is a single configurable value; anything above needs a human decision.
- ◆ **Nothing waits forever** — a pending approval automatically resolves to “No Response” after 3 days.
- ◆ **Workflow-level failure alerting** — a dedicated Error Trigger notifies the team in Slack if the automation itself breaks.
- ◆ **Resilient extraction** — the Gemini request retries automatically on transient failures.

7 Setup Checklist

- ☒ Connect credentials for Gmail, Google Drive, Google Sheets, Google Gemini and Slack.
- ☒ Create a Google Sheet invoices tab with the required headers and select it in all four Sheets nodes.
- ☒ Choose your Drive folder and Slack channel in the relevant nodes.
- ☒ For in-Slack approvals: create a Slack app with chat:write, enable Interactivity, invite the bot to the channel.
- ☒ Set Error Workflow to this workflow in Settings so On Workflow Error actually fires.

8 Node-by-Node Map

Email Intake	Watch Invoice Emails → Is PDF or Image? → Extract Invoice Data → Mark Email As Read
Drive Intake	Watch Invoice Drive Folder → Download File → Extract Invoice Data → Tag Source
Validate & Dedup	Parse & Validate Extraction → Check Duplicate Invoice → Classify Expense Category
Approval & Ledger	Determine Approval Route → Request Manual Approval → Update Ledger Status

SUMMARY

Designed and built end-to-end in n8n, demonstrating multi-channel document intake, AI extraction with validation, duplicate protection, and human-in-the-loop spend approval.